

Love of Variety, Habit Formation, and Dynamic Advertising

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Roadmap

- 1 Motivation
- 2 Model
- 3 Simulation Design
- 4 Results: Consumer Behavior
- 5 Next Steps

Motivation: Data Privacy, Ad Targeting, and Consumer Agency

- Concerns around data privacy and ad targeting
 - Facebook's 5B FTC settlement
 - Google ad tracking opt-out
- Amazon anti-trust scrutiny
 - 70% of US e-commerce; 40% of all retail
 - Uses consumer data to target ads and build market share
- Consumer agency over data (opt out)
 - Apple iOS 14.5 update: 85% opt-out rate
 - Facebook ad revenue down 10% in Q2 2022

Addressing the Concern

- How can we test the impact of policy interventions surrounding data privacy?
- Structural demand estimation is a powerful tool for policy analysis, but requires a model of consumer behavior and firm incentives

Key Challenge

How do we model consumer behavior in a way that captures the key drivers of consumption dynamics and firms' response?

Why Does Variety Matter?

- Canonical logit models ignore consumer preferences for variety, which are empirically important (e.g., McFadden 1974; Berry, Levinsohn, Pakes 1995)
- These preferences shape consumer switching behavior, which in turn shapes firms' incentives to strategically target ads and promotions
- Understanding the role of variety-seeking is crucial for modeling consumer behavior and evaluating the impact data privacy policies on both consumers and firms

Research Question

How much do consumer tastes for novelty and habit influence consumption dynamics and firms' advertising strategies in a dynamic market?

What This Paper Does

- ① **Model:** Derive a discrete-choice demand system where the utility of a product depends on both deviation from the consumer's experienced characteristic mean $\bar{X}_{it}$ and product quality X_{jt}
- ② **Simulation:** Characterize consumer and firm behavior across preference regimes (β_i, γ_i) and advertising specifications
- ③ **Estimation (coming):** Bring model to Nielsen scanner data to recover structural parameters
- ④ **Counterfactuals (coming):** Study advertising targeting policy and welfare implications

Today's talk: Model + simulation results

Literature Connections

Demand side

- McFadden (1974): MNL foundation
- Berry, Levinsohn, Pakes (1995): RC logit
- Erdem & Keane (1996): Learning dynamics

Advertising / firm side

- Nerlove & Arrow (1962): Goodwill
- Akerberg (2001): Informative ads
- Goldfarb & Tucker (2011): Targeted digital ads

Contribution

I develop a dynamic demand model which parameterizes variety-seeking and use it to study how consumers with different preferences for variety respond to both product entry and dynamic advertising strategies.

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Consumer's Problem

Consumer i chooses product j in period t to maximize:

$$U_{ijt} = \underbrace{\beta_{ij} X_{jt}}_{\text{characteristic component}} + \underbrace{\gamma_i \log(1 + \Sigma_{ijt}^2)}_{\text{novelty component}} + \underbrace{\kappa a_{jt}}_{\text{advertising}} - \underbrace{\alpha p_{jt}}_{\text{price}} + \xi_{ij} + \varepsilon_{ijt}$$

- where $\Sigma_{ijt} = X_{jt} - \theta_{it}$ is the deviation of product j 's characteristic from consumer i 's previously chosen characteristics θ_{it} in period t .
- at the moment, $\theta_{it} = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$, or the mean of previously chosen product characteristics.

Key parameters:

- $\beta_i > 0$: marginal utility of quality
- $\gamma_i > 0$: variety-seeking
- $\kappa > 0$: advertising shifter

Choice probability (MNL):

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}}$$

Follows from $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$

State Variable: History of Choices

History of Choices θ_{it}

$$\theta_{it} = f(\{X_{js}\}_{s < t}) = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$$

The consumer's history, for simplicity, is summarized by the mean of previously chosen product characteristics. Updates each period as the consumer makes a new choice. This is the key state variable linking past behavior to current utility.

Initialization strategy:

- 1 **Initial state:** Simulate $T_{\text{prior}} = 5$ pre-sample periods from $J \in [1, 2, 3, 4, 5]$ to generate an experience-based prior. Similar to Bronnenberg, Dubé, Joo (2021).

Initial history yields more heterogeneous behavior — more on this shortly.

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Simulation Parameters

Table: Parameters

Param.	Value	Description
T	100	Periods
J	5	Products
S	1,000	Simulations
κ	0	Advertising (off)
α	0	Price (off)
ξ_j	0	Quality (off)
β_{df}	0	Discount factor (off)

Preference for Characteristic j :

- $\beta_i \in \{1, 2.5, 4\}$ — low, medium, high quality sensitivity

Preference for Variety:

- $\gamma_i \in \{0.5, 1, 1.5\}$ — low, medium, high variety-seeking

$9 \text{ regimes} \times 2 \text{ mean specs} \times 2 \text{ ad specs}$
 $= 36 \text{ cells}$

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Naive Measurement

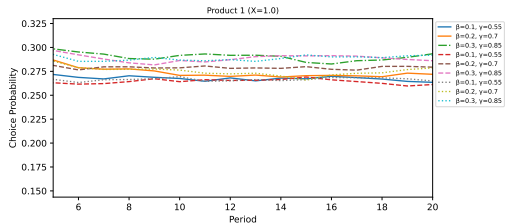
- Near static choice probabilities
- Any variation due to ε_{ijt} , not consumer preferences
- CCPs are not informative about consumer behavior or firm incentives
- Nearly all mass on Product 5, which has the highest quality X_{jt}

Table: Naive CCPs

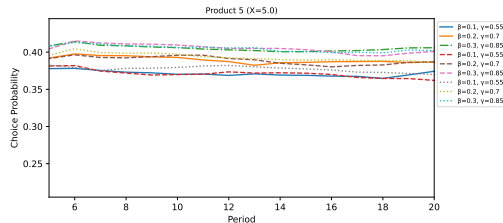
Product	CCP
Product 1	0.02%
Product 2	0.90%
Product 3	5.00%
Product 4	20.0%
Product 5	78.0%

LOV Specification

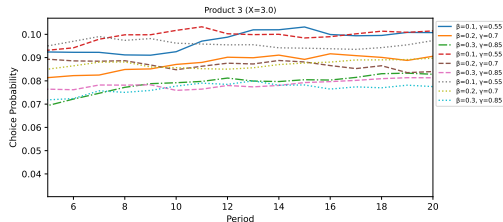
Choice Probability by Product and Regime



Choice Probability by Product and Regime



Choice Probability by Product and Regime



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Road to Estimation

① Multiple consumers

Extend simulation to $I = 5$ consumers who have heterogeneous (β_i, γ_i) ; aggregate to market shares

② Heterogeneous γ_i

Draw $\gamma_i \sim N(0, \sigma_\gamma)$; simulate over $\sigma_\gamma \in \{0.5, 1.0, 1.5\}$

③ Consumer value function

Move from reduced-form simulation to solving the full dynamic discrete choice with continuation values

④ Nielsen scanner data

Match simulated moments to data; identify (β_i, γ_i) using BLP instruments

⑤ Firm's Problem

Ad targeting policy; profit maximization; counterfactuals

Summary

- ① **Model:** Consumer utility depends on deviation from experienced mean; nests variety-seeking and habit formation
- ② **Set mean:** Flat choice probabilities — good benchmark, insufficient for data
- ③ **Prior mean:** Rich behavioral dynamics; regime differences generate cross-sectional sorting consistent with scanner data
- ④ **Firm:** Markovian advertising incentive rises when consumer drifts; generates time-varying promotional intensity
- ⑤ **Next:** Multiple consumers, structural estimation, Nielsen data

Thank you — questions welcome

Appendix

Simulation Algorithm (Consumer)

- ➊ Draw $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$ for all i, j, t
- ➋ Initialize $\bar{X}_{i0}$ (set mean or prior mean)
- ➌ For $t = 1, \dots, T$:
 - ➊ Compute $U_{ijt} = \beta_i X_{jt} - \gamma_i (X_{jt} - \bar{X}_{it})^2 + \kappa a_{jt} + \varepsilon_{ijt}$
 - ➋ Set $j^* = \arg \max_j U_{ijt}$
 - ➌ Update $\bar{X}_{it+1} = \frac{1}{t} \sum_{k \leq t} x_{ijk^*}$
- ➍ Compute CCPs via log-sum-exp for numerical stability:
 $\log P_{ijt} = U_{ijt} - \log \sum_k e^{U_{ikt}}$
- ➎ Average over $S = 1,000$ simulation draws